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Clinical Sensitivity and Specificity of the PROSTest in an American Cohort

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ABSTRACT

Introduction: Prostate cancer (PCa) is the most commonly diagnosed cancer in men in the United States, following skin cancer, with an incidence rate of 112.7 per 100,000 men per year. The need for a reliable, non-invasive diagnostic tool for early PCa detection (screening, biochemical residual disease) remains unmet due to the limitations of PSA testing, which often leads to overdiagnosis and overtreatment. The PROSTest is a novel, blood-based qPCR assay that assesses gene expression to diagnose PCa and predict patient outcomes to different treatments. This study aimed to validate the sensitivity and specificity of the PROSTest in a diverse cohort of US-based PCa patients compared to healthy controls.

Materials and Methods: This prospective study included 143 PCa patients and 92 healthy controls. Blood samples were collected, and the PROSTest was conducted following RNA isolation and cDNA production, using a predefined 27-gene algorithm to provide a binary output. The assay's sensitivity and specificity were evaluated using receiver operating characteristic (ROC) analysis, with a 50% score cut-off distinguishing PCa from non-PCa patients. Analytical reproducibility was assessed with intra- and inter-assay comparisons of Ct values and PROSTest scores.

Results: The PROSTest demonstrated a sensitivity of 94% (95% CI 89–98%) and a specificity of 88% (95% CI 80–94%) in distinguishing PCa patients from controls, with an area under the ROC curve (AUROC) of 0.97. The false positive rate among controls was 12%. Intra- and inter-assay reproducibility was confirmed with no significant differences in Ct values or PROSTest scores between operators or assays. PROSTest scores were significantly higher in PCa patients compared to controls and in those undergoing treatment versus untreated patients.

Conclusion: This validation study confirms the high sensitivity and specificity of the PROSTest in detecting PCa in a diverse USA cohort. The assay's robustness and reproducibility support its potential as a reliable diagnostic tool for PCa detection and monitoring. Further studies are warranted to evaluate its utility across broader populations and treatment settings.

1 | Introduction

In the USA, prostate cancer (PCa) is the most frequently diagnosed cancer in men, excluding skin cancer, and it is the second most common cancer in men worldwide, after lung

cancer. Approximately 248,530 new cases of prostate cancer were diagnosed in the United States in 2021 [1]. The age-adjusted incidence rate is about 112.7 per 100,000 men per year. The lifetime risk of being diagnosed with a prostate cancer is approximately 1 in 8 [1]. About 60% of cases are

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diagnosed in men aged 65 and older, and it is rare in men under 40. African American men have the highest incidence rates of PCa, nearly 1.7 times higher than in Caucasian men [2]. They also have higher mortality rates. Asian American and Hispanic men have lower incidence rates compared to Caucasian men.

The risk of developing prostate cancer is influenced by a combination of genetic, demographic, environmental, and lifestyle factors [3]. Men most at risk for developing prostate cancer are older (especially over 50), African American, and those with a family history of the disease or certain genetic mutations, for example, *BRCA1/BRCA2* and the Lynch syndrome (hereditary non-polyposis colorectal cancer [HNPCC]) and lifestyle factors such as diet, obesity, smoking, environmental exposures and hormonal levels.

Despite considerable advances in diagnosis and treatment, significant limitations remain. PSA testing has played a significant role in the detection of PCas [4]. However, its use remains controversial due to concerns about overdiagnosis and subsequent overtreatment, in addition to its high biological and analytical variabilities [5–7]. A definitive diagnosis requires a prostate biopsy, with histopathological examination to confirm malignancy and assess the Gleason score, which indicates tumor aggressiveness.

Major areas that require focus include early diagnosis of PCa, appropriate identification of individuals who require biopsy, and patient surveillance [8]. A critical requirement is a non-invasive tool that is effective in both diagnosis and assessment of disease status [9]. Prognosis depends on the stage, Gleason score, PSA levels, and response to treatment (<https://www.auanet.org/guidelines-and-quality/guidelines/oncology-guidelines>). Localized PCa has a good prognosis, while advanced or metastatic disease has a poorer outlook. The 5-year relative survival rate for prostate cancer is nearly 100% for localized and regional stages. Even for metastatic disease, the 5-year survival rate is around 31% [10, 11]. Localized PCa is manageable, but a validated non-invasive, relatively inexpensive tool is still an unmet need.

The PROSTest is a novel biomarker encompassing 27 mRNA gene expressions associated with PCa biology, in addition to 3 house-keeping genes. It is a qPCR test that utilizes RNA isolated from whole blood. Using a proprietary algorithm, the PROSTest provides a binary output for PCa detection with a clinically validated cut-off of 50 (on a 0–100 scale) [12]. Elevated PROSTest scores have been reported to correlate with a poorer overall survival [13] and development of MRD (minimal residual disease) [14] while changes in score have been associated with treatment response to a variety of different ligands (ADT, chemotherapy and radioligand therapy [RLT]).

The PROSTest is associated with encouraging sensitivity and specificity (> 90%) in PCas encompassing the spectrum of disease (adenocarcinoma to metastatic castration-resistant disease) [12] and including men of African genetic heritage [15]. Clinical studies conducted thus far were in Europe and Barbados; this study focused on a diverse American (USA) cohort. This study aimed to validate the sensitivity and specificity of the PROSTest in a diverse cohort of US-based PCa patients compared to healthy controls.

2 | Materials and Methods

2.1 | Patient Selection and Ethics Approvals

This study was a prospective blood collection study comprised of two cohorts including 143 PCa and 92 controls. All subjects were male. The study protocol (WIRB20191743) was approved by the Western Institutional Review/WCG Board and conducted in accordance with Good Clinical Practice principles. Informed consent was obtained from all study participants.

Patients were eligible for the PCa cohort if they had histologically confirmed PCa, had disease documented on a diagnostic scan, were off cytotoxic chemotherapy or radiopharmaceutical therapy (RPT) for at least 4 weeks prior to date of blood collection, and had no other active malignancy within 3 years of enrollment, with the exception of adequately treated basal cell or squamous cell skin cancer. Controls were healthy subjects or patients with no current histologically or cytologically proven diagnosis of a malignancy.

2.2 | Blood Collection

After informed consent was obtained and eligibility of subjects confirmed, a single blood sample was collected per patient. Blood samples were collected in Wren's proprietary RNA stabilization buffer tubes per the sample collection protocol. Specimens were deidentified and coded prior to shipment for analysis. Samples were stored at -80°C until analyzed in batches by Wren Laboratories. Data was collected on the standard-of-care PSA for patients with PCa who had tumor markers drawn at or within a month of the PROSTest blood sample collection.

2.3 | PROSTest Measurements

Randomly selected, coded samples were sent to Wren Laboratories (CAP#8640840), Connecticut, USA for blinded PROSTest measurement. The latter comprises a 2-step protocol (RNA isolation/RT-qPCR) [16]. Transcripts (mRNA) were isolated from whole blood (Mini Blood kit, Qiagen, Valencia, CA), and real-time qPCR was performed on pre-spotted PCR plates (Life Technologies, Carlsbad, CA) [12]. Target transcript levels were normalized to *ALG9*, *TOX4* and *TPT1* and quantified using $\Delta\Delta C_t$ [12]. Final results are expressed as a risk stratification based on a 0–100 score for PCa. The cut-off is 50% for risk of disease [12].

2.4 | Assay Analytics

A total of 30 paired clinical samples including PCa samples ($n = 20$) and controls ($n = 10$) were evaluated by two independent operators using two different PCR machines over a period of 20 days. C_t values and PROSTest scores were evaluated for intra- and inter-assay metrics per CLSI EP05-A3 guidelines (<https://clsi.org/standards/products/method-evaluation/documents/ep05/>) [17].

2.5 | Statistical Considerations and Sample Size Calculation

The primary objective of this protocol was to determine the analytical performance criteria (sensitivity and specificity) of the PROSTest in clinical samples. The study was designed to test the hypotheses that the PROSTest has a sensitivity and specificity of 90% or more in PCa patients. The sample size calculation was based on the assumption that a sensitivity and specificity of greater than 90% would generate further interest in the test for unselected PCa patients. Power and type 1 error were 90% and 2.5%, respectively. Under this model, 90 or more positive tests in a cohort of 100 PCa patients would lead to the rejection of the null hypothesis, suggesting that the PROSTest is sensitive. Likewise, 80 or more negative tests in 100 non-PCa patients would suggest that the PROSTest is specific.

Descriptive statistics were used for patient demographics. Receiver operating characteristics (ROC) curve analysis was used to assess the performance metrics of the PROSTest. Exact 95% confidence intervals (CIs) were calculated for each proportion of interest. Data are provided as median (range) and [mean $\pm$ SD]. PRISM9.4.0 for Windows (GraphPad Software, La Jolla, California, USA, www.graphpad.com), and MedCalc Statistical Software version 20.109 (MedCalc Software bvba, Ostend, Belgium; <http://www.medcalc.org>; 2017) were utilized. Statistical significance was defined as $p < 0.05$.

3 | Results

3.1 | Assay Analytics

We first evaluated the assay reproducibility focusing on both evaluating target Ct values (PCR output – from each gene) as well as PROSTest scores (algorithmic output – based on all genes). An evaluation of Ct values in the intra-assay evaluation ($n = 30$ pairs) identified no significant difference in mean Ct values (FOP: 36 ± 1.1 vs. SOP: 36 ± 1.6 , $p = 0.23$,

Mann–Whitney test; Figure 1A). Ct values were well-correlated (Pearson $r = 0.83$, $p < 0.0001$). In the inter-assay evaluation ($n = 30$ pairs), Ct values were similarly not different (FOP: 36 ± 1.0 vs. SOP: 36 ± 1.1 , $p = 0.09$, Mann–Whitney test; Figure 1B). The calculated intra- and inter-assay variations for Ct values were $0.83 \pm 1.2\%$ and $1.2 \pm 1.1\%$, respectively.

An evaluation of PROSTest scores (Figure 2A) in the intra-assay evaluation ($n = 30$ pairs) identified no significant difference in mean scores (FOP: 61 ± 30 vs. SOP: 60 ± 31 , $p = 0.62$, Mann–Whitney test; Figure 2B). PROSTest scores were well-correlated (Pearson $r = 0.97$, $p < 0.0001$, Figure 2C). In the inter-assay evaluation ($n = 30$ pairs), PROSTest scores were similarly not different (FOP: 60 ± 31 vs. SOP: 60 ± 31 , $p = 0.80$, Mann–Whitney test; Figure 2D). PROSTest scores were well-correlated between the two operators (Pearson $r = 0.83$, $p < 0.0001$, Figure 2E).

3.2 | Clinical Sample Evaluation

The demographics for the patients and controls evaluated are included in Table 1.

In the PCa cohort, the age distribution range was 44–98 years with a mean value of 70. In the control cohort, the age range was 49–79 with a mean value of 59. PROSTest scores for all cohorts are depicted in Figure 3.

3.2.1 | Assay Metrics

In the PCa cohort ($n = 143$), the PROSTest score was 82 (15–97) [79 ± 16] (Figure 3A). In the control cohort ($n = 92$), PROSTest scores were 8 (1–80) [17 ± 22]. Levels were significantly higher in PCas ($p < 0.0001$).

Using ROC curve analyses, the AUROC for differentiating PCa from controls was 0.968 ± 0.01 (Figure 3B). Using the 50% score

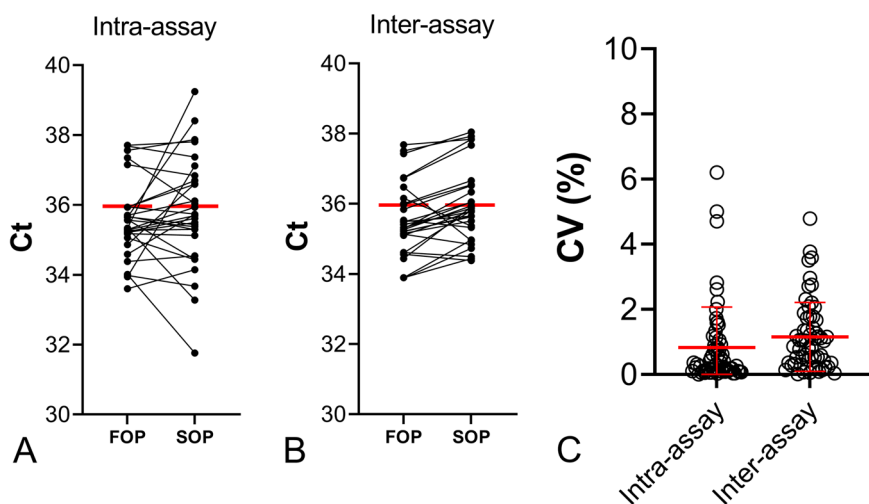


FIGURE 1 | Ct levels: analytical evaluations. (A) Intra-assay reproducibility between the first operator (FOP) and second operator (SOP). Red line is mean value ($N = 30$ /operator). (B) Inter-assay reproducibility between the first operator (FOP) and second operator (SOP). Red line is mean value ($N = 30$ /operator). (C) Intra- and inter-assay coefficient of variation (CV) ($N = 60$) Mean $\pm$ SD. [Color figure can be viewed at wileyonlinelibrary.com]

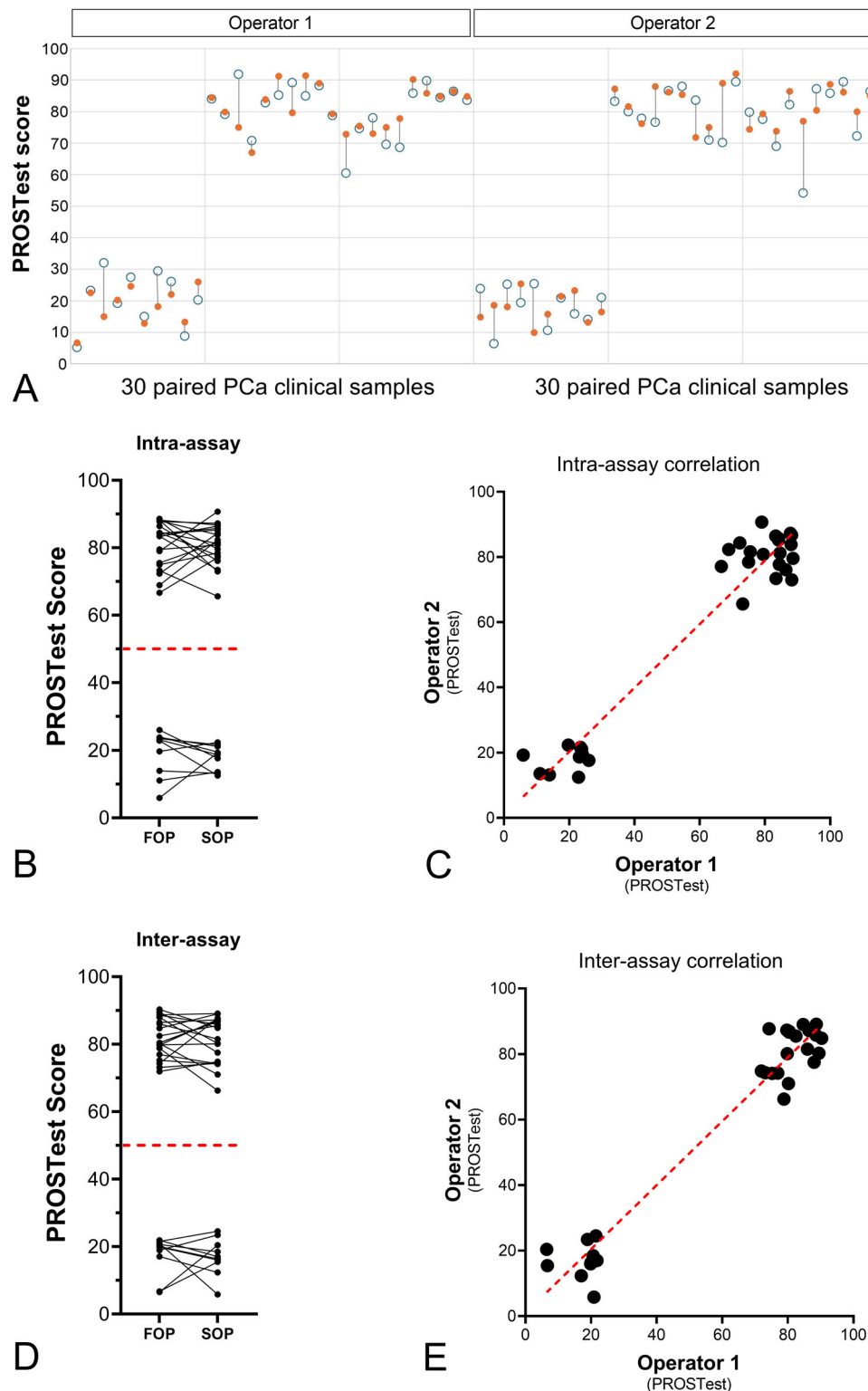


FIGURE 2 | PROTest scores: analytical evaluations. (A) PROTest scores by operator and sample ($n = 30$ pairs/operator). The vertical line connects the two replicates. (B) Intra-assay reproducibility between the first operator (FOP) and second operator (SOP). Red dotted line is mean value ($N = 30$ /operator). (C) Intra-assay correlation. Red dotted line is regression line ($N = 30$ /operator). (D) Inter-assay reproducibility between the first operator (FOP) and second operator (SOP). Red dotted line is mean value ($N = 30$ /operator). (E) Inter-assay correlation. Red dotted line is regression line ($N = 30$ /operator). The red dashed line in Figure 3B,D is the 50% cut point. [Color figure can be viewed at wileyonlinelibrary.com]

cutoff, the PROTest sensitivity was 94% (95% CI, 89%–98%) and specificity was 88% (80%–94%). Our planned analysis identified 11 positives out of 92 controls. This 12% positivity in normal can be false positives or some of the subjects may have

occult disease that has not yet been detected. The result meets our predefined criterion of <20% using the 50% cutoff. This confirmed that the assay had value for detecting PCAs and differentiating patients with the disease from controls.

TABLE 1 | Demographics of the PCa and control cohorts.

	Controls (<i>n</i> = 92)	PCa (<i>n</i> = 143)
Age (years)	59 (49–79)	70 (44–98)
Ethnicity (A:B:H:W)	1:20:31:40	2:25:0:116
PSA (ng/mL)	N/A	4.8 (0.03–42)
PROSTest scores	8 (1–80)	82 (15–97)

PCa = prostate cancer; N/A = not applicable; Ethnicity: A = Asian; B = Black; H = Hispanic; W = White.
Values are given as median (range).

3.3 | Further Characterization of the PROSTest in the PCa Cohort

We further evaluated the value of the PROSTest in the PCa group. An examination of age identified no relationship between PROSTest positivity and years (by decade: $\chi^2 = 2.97$, $p = 0.396$). In terms of ethnicity, no differences ($\chi^2 = 2.42$, $p = 0.298$) in PROSTest positivity were identified between Asians (2/2 [100%]), African Americans (22/25 [88%]), and Whites (111/116 [96%]) with PCa.

The PCa cohort was diverse and comprised 3 distinct groups: a post-treatment cohort ($n = 54$), a cohort undergoing treatment ($n = 26$) and those never treated ($n = 63$). Demographics are included in Table 2.

PSA results were available in 118 (80%) and multiple PSAs were available in 68, for whom we calculated PSA doubling times and PSA velocities. Individual PROSTest scores and PSA levels are included in Figure 4A,B. PROSTest scores were elevated in those undergoing treatment compared to untreated ($p = 0.008$), but there were no differences in the proportion of positive cases. PSA levels were elevated in the never treated cohort ($p < 0.0001$) versus those on treatment.

3.3.1 | PROSTest in the Post-Treatment Group ($n = 54$)

Four (7%) were negative. These all had prostatectomy and radiation therapy (SBRT: $n = 2$, EBRT $n = 2$). PROSTest scores were not correlated with PSA levels ($r = +0.11$, $p = 0.46$) or statistically different between individuals with PSA < 0.5 ng/mL ($n = 25$: 78 ± 18), 0.5 – 1 ng/mL ($n = 10$: 75 ± 15) or > 1 ng/mL ($n = 19$: 82 ± 14). PSADT and velocities were evaluable in 33 individuals. Significantly higher PROSTest scores ($p = 0.0025$) were noted in those with a positive PSA velocity/PSA doubling time (Figure 4C).

3.3.2 | PROSTest in the Treatment Group ($n = 26$)

One (4%) of 26 was negative. This was an individual (72 White, G5, lumbar metastases) being treated with Eligard (3 years) and who had received 30 Gy EBRT (prior to hormonal therapy). PROSTest scores were not correlated with PSA levels ($r = +0.03$, $p = 0.9$) or were statistically different between individuals with PSA < 0.5 ng/mL ($n = 13$: 82 ± 18) or > 1 ng/mL ($n = 8$: 85 ± 9).

PSADT and velocities were evaluable in 12 individuals. No relationships were identified between PROSTest scores and these parameters.

3.3.3 | PROSTest in the Never Treated Group ($n = 63$)

This comprised 25 individuals in active surveillance (AS, time since diagnosis: 21 months [7–63]) and 38 newly diagnosed (ND, time since diagnosis: 1.5 months [0–6]). None had undergone any form of treatment. All had biopsy confirmation of the disease.

3.3.4 | PROSTest in AS ($n = 25$)

Two of 25 (8%) had negative scores. One was a 62-year White individual, G1, T1N0M0, 16 months after diagnosis with a PSADT of -53.6 months and PSA velocity of -1 ng/mL/year. The second was a 69-year Black individual, Grade Group 1 (G1: Gleason score ≤ 6), T1N0M0, 56 months after diagnosis. Multiple PSAs were not available to calculate PSADT or the PSA velocity. The majority of men in this cohort were G1 ($n = 22$) or G2 (Grade Group 2: Gleason score 7 (3 + 4), $n = 3$) and were staged as T1a ($n = 7$)/T1c ($n = 16$) or T2a ($n = 2$). None had metastatic disease. Significantly higher PROSTest scores were noted in those with G2 ($p = 0.027$) and T2a tumors ($p = 0.0047$) as well as those with a positive PSA velocity/PSA doubling time ($p = 0.01$) (Figure 5).

3.3.5 | PROSTest in ND ($n = 38$)

One of 38 (3%) had a negative score. This was a 57-year White individual, G3 (Grade Group 3 (Gleason Score 7 [4 + 3]), pT3a. Scores were not significantly associated with Grade or stage in this cohort, but the evaluable subjects in each of the groups were small ($n = 1$ – 5) and insufficient for analysis.

4 | Discussion

In a US-based validation study of the PROSTest, the test demonstrated strong diagnostic performance for detecting PCa across a diverse cohort (age, ethnicity and disease subgroups). This study, which included 143 PCa patients and 92 controls, highlights the PROSTest's clinical utility in diagnosing PCa, monitoring treatment responses, and identifying individuals with aggressive disease. The test, a qPCR-based assay using a 27-gene algorithm, exhibited a sensitivity of 94% and specificity of 88% (using a cut-off of 50) in distinguishing PCa patients from healthy individuals. With an AUROC curve of 0.97, the PROSTest's diagnostic accuracy is supported in real-world settings. Out of 92 controls, 11 false positives were identified, resulting in a false positive rate of 12%, which met the pre-defined protocol criterion of less than 20%. These findings support previous research conducted predominantly in Afro-Caribbean men, showing that the PROSTest outperforms PSA as a more sensitive and specific biomarker for diagnosing and potentially screening for prostate cancer [15].

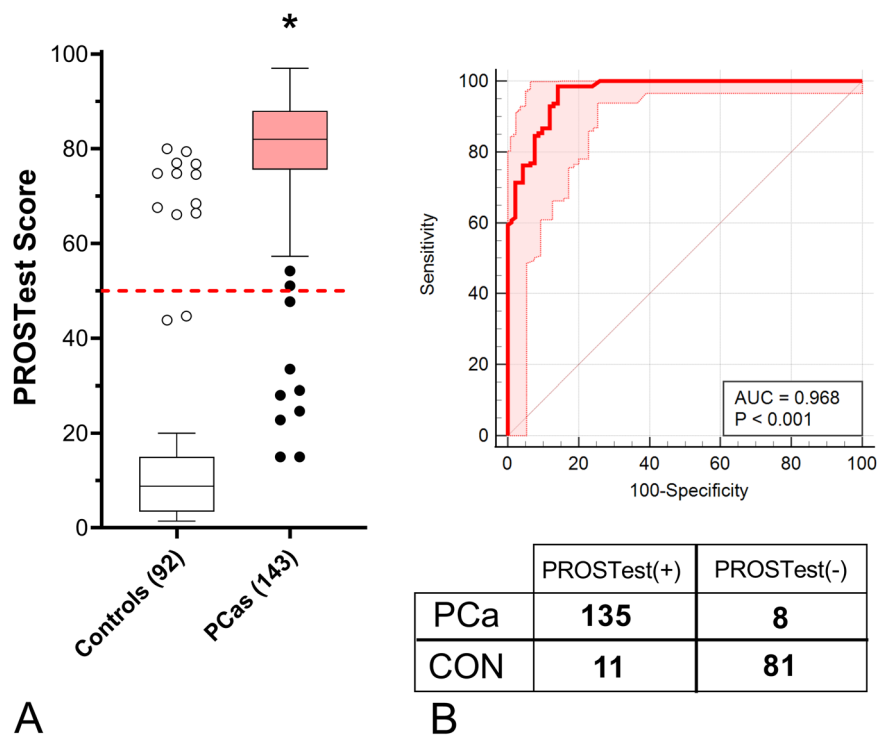


FIGURE 3 | PROTest evaluation analysis. (A) Tukey plots of PCa probabilities in whole blood from controls ($n = 92$) and PCas ($n = 143$). The red dashed line is the 50% cut point. (B) Area under the receiver operating characteristic (AUROC) curve for differentiation of PCa from controls was identified as 0.97. The data for 95% confidence interval of the AUROC line is highlighted. [Color figure can be viewed at wileyonlinelibrary.com]

PSA screening, a traditional method used for decades, has shown mixed results regarding its effectiveness in reducing prostate cancer mortality. While trials in both the United Kingdom [18] and the United States [19] found no significant reduction in mortality, a European trial [20] demonstrated a notable decrease in prostate cancer deaths. However, the European trial was also associated with a higher false positive rate of 17.8%, leading to concerns about unnecessary biopsies and overdiagnosis [21]. Overdiagnosis rates from PSA screening have been reported to range from 20.7% to 50.4% [7], and its overall impact on reducing prostate cancer mortality remains a topic of debate. Additionally, the risks associated with the outcomes of PSA screening, such as biopsy complications and overtreatment, contribute to its controversial status [7].

Circulating tumor DNA (ctDNA) and exosome-based tests, while valuable in certain contexts, fall short in several key areas. ctDNA analysis, which tracks tumor-derived DNA fragments in the bloodstream, lacks sensitivity in early-stage or localized prostate cancer because of the low levels of tumor DNA present [9]. This limitation makes ctDNA less reliable for early detection, though its utility increases in advanced stages where tumor burden is higher. Moreover, ctDNA testing often requires complex technologies like whole-exome sequencing or digital PCR, adding complexity and cost to the process. Exosome-based tests, which analyze urinary exosomal markers such as PCA3 and PSMA, also face challenges in diagnostic accuracy. These tests generally yield lower AUROC values (ranging from 0.731 to 0.870) compared to the PROTest [22]. Even when combined with serum PSA, the highest AUROC obtained was 0.914 [22], which is still lower than the 0.968 achieved by the PROTest. Overall, the diagnostic limitation of

these nascent technologies, particularly in early detection, hinder their broader clinical application [23]. In contrast, the PROTest's qPCR-based approach offers well-validated superiority in accuracy, simplicity, and reliability, making it a more comprehensive solution for early detection, treatment monitoring, and guiding clinical decisions in PCa.

In a cohort of 25 men undergoing AS, the PROTest score showed promising potential as a monitoring tool for identifying higher-grade and more aggressive prostate cancer (see Figure 5). Of the 25 AS patients, only two (8%) returned negative PROTest scores. These individuals were both low-grade (Grade Group 1 [Gleason 1]) and early-stage (T1N0M0), with no available PSA doubling time or velocity for one patient. The majority of the cohort was Grade Group 1 ($n = 22$), with three patients classified as Grade Group 2 [Gleason Score 7 (3 + 4)], and staged primarily as T1a/T1c. None had metastatic disease. Significantly higher PROTest scores were observed in patients with Grade Group 2 tumors ($p = 0.027$) and stage T2a tumors ($p = 0.0047$). Moreover, those with a positive PSA velocity or doubling time had markedly higher PROTest scores ($p = 0.01$), indicating a correlation between higher test scores and more aggressive disease progression. These findings suggest that the PROTest score may be valuable for monitoring patients in AS, offering insight into tumor grade and progression risk, although this requires larger numbers and prospective validation.

A key limitation of this study is its relatively small sample size, particularly within the AS cohort and other subgroups. With only 25 AS patients and a limited representation of more advanced disease stages, the findings may not be fully generalizable to broader populations of PCa patients, especially those

TABLE 2 | Demographics of the PCa cohort.

	Post-treatment ^a (n = 54)	Treated ^b (n = 26)	Never treated (n = 68)	
			AS (n = 25)	ND (n = 38)
Age (years)	73 (56–83)	72 (44–98)	72 (59–81)	67 (50–86)
Time since diagnosis^c	48 (6–165)	18 (1–176)	21 (7–63)	1.5 (0–6)
Ethnicity (A:B:H:W)	0:4:0:50	2:0:0:24	0:9:0:16	0:12:0:26
Gleason Grade Group				
G1	31 (57%)	3 (11%)	22 (88%)	12 (32%)
G2	9 (17%)	1 (4%)	3 (12%)	6 (16%)
G3	5 (9%)	5 (19%)	0 (0%)	8 (20%)
G4	7 (13%)	7 (27%)	0 (0%)	6 (16%)
G5	2 (4%)	3 (12%)	0 (0%)	5 (13%)
N/A	—	7 (27%)	—	1 (3%)
T-stage				
T1	48 (89%)	12 (46%)	23 (92%)	24 (63)
T2	6 (11%)	4 (15%)	2 (8%)	5 (13%)
T3	0 (0%)	2 (8%)	0 (0%)	4 (11%)
T4	0 (0%)	0 (0%)	0 (0%)	0 (0%)
N/A	—	8 (31%)	—	5 (13%)
N-stage				
N0	50 (93%)	14 (54%)	25 (100%)	18 (47%)
N1	0 (0%)	4 (15%)	0 (0%)	2 (6%)
N/A	4 (7%)	8 (31%)	—	18 (47%)
M-stage				
M0	49 (91%)	16 (62%)	25 (100%)	17 (45%)
M1	1 (2%)	8 (31%)	0 (0%)	—
N/A	4 (7%)	2 (7%)	—	21 (55%)
PSA (ng/mL)	0.49 (0.03–19)	0.40 (0.03–42)	6.8 (0.11–12)	7.9 (3.7–31)
PSADT	−3.7 ^d (−66 to 59)	−0.90 ^e (−18 to 9.3)	14 ^f (−54 to +314)	13 ^g (−62 to +68)
PSA velocity	−0.40 ^d (−21 to 3.1)	−22 ^e (−522 to 15)	0.20 ^f (−14 to +100)	2.0 ^g (−0.9 to +74)
PROTest scores	83 (15–96)	87 (23–97)	79 (15–94)	80 (25–94)

Abbreviations: AS = active surveillance; EBRT = external beam radiation therapy; IGRT = image-guided radiation therapy; IMRT = intensity modulated radiation therapy; N/A = not available; ND = newly diagnosed; SBRT = stereotactic body radiation therapy.

^aNot in current treatment. Previous treatment included IMRT (n = 34: 50.5–81 Gy), EBRT (n = 3: 70–81 Gy), SBRT (n = 3, 36.25 Gy), surgery (n = 11), Lupron (n = 3).

^bCurrent therapies include EBRT (n = 4), IGRT (n = 3), IMRT (n = 12), Abiraterone (n = 2), Lupron (n = 3), Firmagon (n = 2).

^cMonths

^dn = 33/54 calculatable.

^en = 12/26 calculatable.

^fn = 15/24 calculatable.

^gn = 8/38 calculatable.

with more aggressive or metastatic disease. While the study shows promising results in distinguishing PCa patients from healthy controls, larger, more diverse cohorts are needed to validate the performance of the PROTest across different patient populations, disease stages, and treatment regimens. Additionally, the small sample size may limit the statistical power to detect subtle differences in test performance and correlations with clinical outcomes. Prospective, multicenter studies involving larger cohorts are essential to confirm these findings and fully establish the utility of the PROTest as a diagnostic and monitoring tool.

5 | Conclusion

This US validation study identifies that the sensitivity of the PROTest is exceptionally high in a heterogeneous population of PCas including men who have undergone treatment, in those undergoing treatment, individuals in active surveillance programs and those newly diagnosed. Specificity within a healthy population of participants is also exceptionally high. This confirms two prior studies; one undertaken in Caucasian men from two European cohorts (Germany and Poland) [12], the second in men of African genetic heritage (Barbados) [15]. The

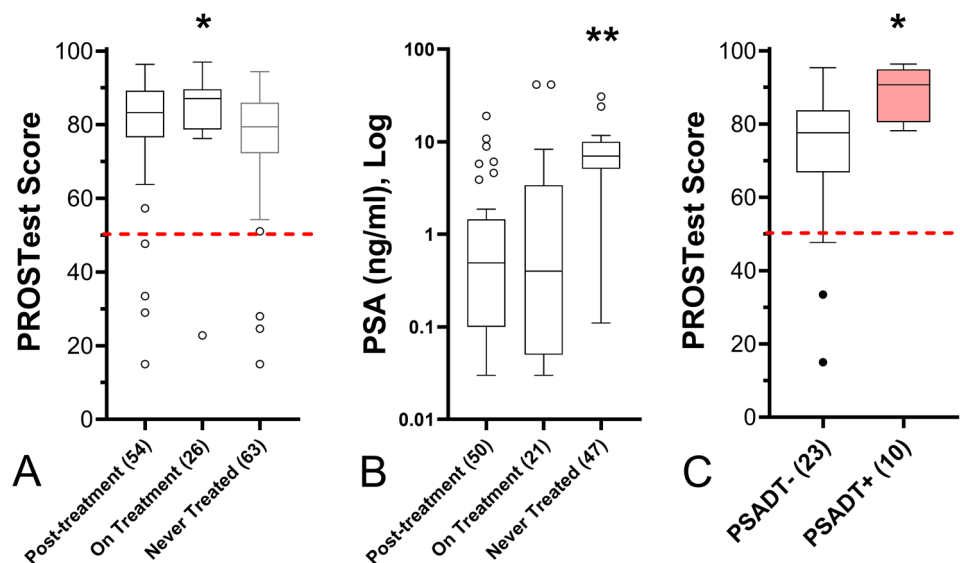


FIGURE 4 | Plots of PROTest scores and PSA levels in the PCa cohorts. (A) Tukey plot of PROTest scores in each of the 3 cohorts. Scores were significantly ($p = 0.008$) elevated in the subjects on treatment. The red dashed line is the 50% cut point. (B) Tukey plot of PSA levels in each of the 3 cohorts. Scores were significantly ($p < 0.0001$) elevated in the subjects who were never treated. (C) Tukey plot of PROTest scores in the post-treatment cohort. Scores were significantly ($p = 0.0025$) elevated in individuals with a positive PSA doubling time (PSADT+). The red dashed line is the 50% cut point. [Color figure can be viewed at wileyonlinelibrary.com]

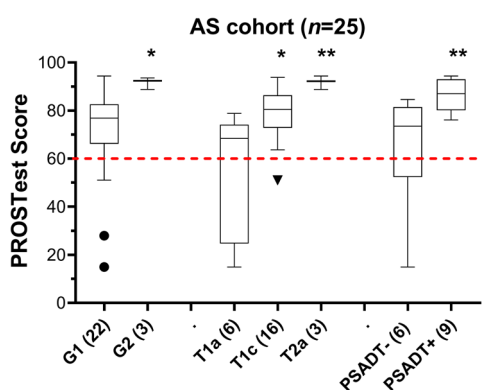


FIGURE 5 | Relationship between PROTest scores and grade, staging, and PSA doubling times (PSADT) in the Active Surveillance (AS) cohort. Tukey plots of PROTest scores versus grade, stage and doubling time. * $p < 0.05$. ** $p \leq 0.01$. The red dashed line is the 50% cut point. Outliers are identifiable by black circles or inverted triangles. G1 = Grade Group 1 (Gleason Score ≤ 6). G2 = Grade Group 2 (Gleason Score 7 [3 + 4]). [Color figure can be viewed at wileyonlinelibrary.com]

evaluation in the AS cohort identifies the test scores may have value in identifying more aggressive disease. Larger, prospective studies are warranted to confirm the value of PROTest across diverse PCa patient populations and different treatment regimens (including AS) in the United States.

Author Contributions

Conception and design: Abdel Halim, Mark Kidd, Ignat Drozdov, R. David Rosin, Kambiz Rahbar. **data analysis and interpretation:** Ignat Drozdov, Mark Kidd, Abdel Halim, Vikas Prasad. **data acquisition:** Kambiz Rahbar, Mark Kidd, Ignat Drozdov, Abdel Halim. **Critical revision of the manuscript for scientific and factual content:** Abdel Halim, Mark Kidd, Ignat Drozdov, Vikas Prasad, R. David Rosin, Kambiz

Rahbar. **Drafting the manuscript:** Abdel Halim, Mark Kidd, Ignat Drozdov, Vikas Prasad, R. David Rosin, Kambiz Rahbar. **Statistical analysis:** Mark Kidd, Ignat Drozdov, Abdel Halim.

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Ethics Statement

All studies were approved by the responsible ethics review board (see Methods). The research conformed to the principles of the Helsinki Declaration (2013).

Consent

Individuals provided personal information and consented to follow-up (See Materials and Methods: Patient Selection and Ethics Approvals).

Conflicts of Interest

Kambiz Rahbar reports consultation/lectureship fees from ABX, AAA, Bayer, AMGEN, SIRTEX and Janssen Cilag. Mark Kidd and Abdel Halim are employees of Wren Laboratories. Ignat Drozdov has consulted for Wren Laboratories and is a shareholder at Bering Limited. R. David Rosin and Vikas Prasad have no conflicts.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. R. L. Siegel, K. D. Miller, H. E. Fuchs, and A. Jemal, "Cancer Statistics, 2021," *CA: A Cancer Journal for Clinicians* 71, no. 1 (2021): 7–33.
2. R. L. Siegel, K. D. Miller, and A. Jemal, "Cancer Statistics, 2019," *CA: A Cancer Journal for Clinicians* 69, no. 1 (2019): 7–34, <https://doi.org/10.3322/caac.21551>.

3. M. S. Pagadala, A. Lui, J. Lynch, et al., "Healthy Lifestyle and Prostate Cancer Risk in the Million Veteran Program," *Cancer* 130 (2024): 3496–3505.
4. J. E. Oesterling, "Prostate Specific Antigen: A Critical Assessment of the Most Useful Tumor Marker for Adenocarcinoma of the Prostate," *Journal of Urology* 145, no. 5 (1991): 907–923.
5. J. J. Fenton, M. S. Weyrich, S. Durbin, Y. Liu, H. Bang, and J. U. S. Melnikow, "Preventive Services Task Force Evidence Syntheses, formerly Systematic Evidence Reviews," in *Prostate-Specific Antigen-Based Screening for Prostate Cancer: A Systematic Evidence Review for the U.S. Preventive Services Task Force* (Rockville, MD: Agency for Healthcare Research and Quality (US), 2018).
6. A. Maeda-Minami, T. Nishikawa, H. Ishikawa, et al., "Association of PSA Variability With Prostate Cancer Development Using Large-Scale Medical Information Data: A Retrospective Cohort Study," *Genes and Environment* 45, no. 1 (2023): 25.
7. J. J. Fenton, M. S. Weyrich, S. Durbin, Y. Liu, H. Bang, and J. Melnikow, "Prostate-Specific Antigen-Based Screening for Prostate Cancer: Evidence Report and Systematic Review for the US Preventive Services Task Force," *Journal of the American Medical Association* 319, no. 18 (2018): 1914–1931.
8. S. Gillessen, G. Attard, T. M. Beer, et al., "Management of Patients With Advanced Prostate Cancer: Report of the Advanced Prostate Cancer Consensus Conference 2019," *European Urology* 77, no. 4 (2020): 508–547.
9. I. Casanova-Salas, A. Athie, P. C. Boutros, et al., "Quantitative and Qualitative Analysis of Blood-Based Liquid Biopsies to Inform Clinical Decision-Making in Prostate Cancer," *European Urology* 79, no. 6 (2021): 762–771.
10. W. T. Lowrance, R. H. Breau, R. Chou, et al., "Advanced Prostate Cancer: AUA/ASTRO/SUO Guideline Part I," *Journal of Urology* 205, no. 1 (2021): 14–21.
11. W. T. Lowrance, R. H. Breau, R. Chou, et al., "Advanced Prostate Cancer: AUA/ASTRO/SUO Guideline Part II," *Journal of Urology* 205, no. 1 (2021): 22–29.
12. I. M. Modlin, M. Kidd, I. A. Drozdov, et al., "Development of a Multigenomic Liquid Biopsy (PROSTest) for Prostate Cancer in Whole Blood," *Prostate* 84, no. 9 (2024): 850–865.
13. A. Chirindel, M. Kidd, C. A. Rentsch, et al., "Evaluation of a Novel Transcriptomic Tumor Signature (PROSTest) as Response Biomarker for 177Lu-PSMA Therapy in Advanced Prostate Cancer," *Journal of Clinical Oncology* 41, no. 6 Suppl (2023): 183.
14. K. Rahbar, M. Kidd, K. E. Seitzer, A. J. Schrader, M. Boegemann, and I. M. Modlin, "Prediction of Early Prostate Cancer Recurrence Using a Liquid Biopsy Approach," *Journal of Clinical Oncology* 41, no. S6 (2023): 386.
15. R. D. Rosin, A. Haynes, M. Kidd, I. Drozdov, I. Modlin, and A. Halim, "Evaluation of a Multigenomic Liquid Biopsy (PROSTest) for Prostate Cancer Detection and Follow-Up in a Caribbean Population," *Cancer Epidemiology* 92 (2024): 102642.
16. M. Kidd, I. A. Drozdov, S. Matar, et al., "Utility of a Ready-to-Use PCR System for Neuroendocrine Tumor Diagnosis," *PLoS One* 14, no. 6 (2019): e0218592, eCollection 0212019, <https://doi.org/10.1371/journal.pone.0218592>.
17. A. B. Halim, *Biomarkers, Diagnostics, and Precision Medicine in the Drug Industry: Critical Challenges, Limitations and Roadmaps for Best Practices* (Cambridge, MA: Elsevier, 2019).
18. R. M. Martin, J. L. Donovan, E. L. Turner, et al., "Effect of a Low-Intensity PSA-Based Screening Intervention on Prostate Cancer Mortality: The CAP Randomized Clinical Trial," *Journal of the American Medical Association* 319, no. 9 (2018): 883–895.
19. G. L. Andriole, E. D. Crawford, R. L. Grubb, et al., "Mortality Results From a Randomized Prostate-Cancer Screening Trial," *New England Journal of Medicine* 360, no. 13 (2009): 1310–1319.
20. F. H. Schröder, J. Hugosson, M. J. Roobol, et al., "Screening and Prostate-Cancer Mortality in a Randomized European Study," *New England Journal of Medicine* 360, no. 13 (2009): 1320–1328.
21. T. P. Kilpeläinen, T. L. J. Tammela, M. Roobol, et al., "False-Positive Screening Results in the European Randomized Study of Screening for Prostate Cancer," *European Journal of Cancer* 47, no. 18 (2011): 2698–2705.
22. J. Gan, X. Zeng, X. Wang, et al., "Effective Diagnosis of Prostate Cancer Based on mRNAs From Urinary Exosomes," *Frontiers in Medicine* 9 (2022): 736110.
23. E. Endzeliņš, V. Melne, Z. Kalniņa, et al., "Diagnostic, Prognostic and Predictive Value of Cell-Free miRNAs in Prostate Cancer: A Systematic Review," *Molecular Cancer* 15, no. 1 (2016): 41.